EXPERIMENT - 1

Basic Networking Commands

Aim:

study of various Network Commands used in linux & windows

1) arp -a:

ARP is short form of address resolution Protocol, it will show the Ip address of your computer along with the Ip address and MAC address of your router.

Output:

i) (172.16.9.212) at 7e.57.58.38.26:24 [either] on enp 350
(ii) (172.16.8.249) at 00:27:02:13:15:77 [either] on enp 350
(iii) (172.16.11.207) at 7c:51:58:38:36:87 [either] on enp 350
(iv) (172.16.8.131) at 00:21:0e:13:14:41 [either] on enp 350
(v) (172.16.8.145) at 00:27.0e:13.ed:44 [either] on enp 350

2) hostname

This is the simplest of all TCP/IP Commands of simply displays the name of your computer.

Output:

localhost.localdomain.

3) Netstat

Netstat displays a variety of statics about a computers active TCP/Ip Connection. it is Command line tool for monitoring network Connection both incomming & outgoing as well as viewing routing tables interfaces and non-increment.

output.

Proto	Recv-Q	Send-Q	LA	FA	state.
tcp	0	0	localhost localdrama.	172.16.8. 135.5[illegible]949	Estable
tcp	0	0	localhost localdrama.sh	172.16.8.135 59129	Estable
tcp	0	0	localhost localdrama.sh	172.16.8.11. 59129	Estable.

4) nslookup:

It is a tool used to perform CNS lookups in linux, it is used to display DNS details such as Ip address of a Particular computer. The MX records.

Eg: nslookup www.google.com

output:

Server: 172.16.8.1

address: 172.16.8.1#53

www.snapchat.com Canonical name web.gcb.api.snapchat.com

web-gcp.api.snapchat.com

Name: web-gcp.api.sc.ge.com

address: 34.149.46.130

5) Ping (Packet internet Groper)

Command is the best way to test connection btw two nodes ping use to communicate to each other devices.

1) #ping hostname (ping localhost)

output:

64 bytes from localhost (!!1) : 1 cmp-seq=1 #L=64
time = 0.11ms

64 bytes from localhost(!!1) : 1 cmp-seq 2 #L=64
time = 0.036ms

64 bytes from localhost (!!1) : 1 cmp-seq=3 #L=64
time = 0.035ms

64 bytes from localhost(!!1) : 1 cmp-seq=4 #L=64
time = 0.036ms

6) Router:

Router command is used to show manipulate the Ip raining table. It is primarily setup to static route to specify host.

Output:

Destination	Gateway	Germany	flag	metrics.
default	gateway	0.0.0.0	0.67	100
172.16.8.0	0.0.0.0	255.255.2520	U	100

7) Ip

The Ip Command us one of the basis Command every administrator will need in daily usark, from setting up new system and assigning Ip to trouble the Ip Command can show address information, manipulate routing, plus display network various device & tunnels.

Output:

clink/loopback 00.00.00.00.00.00 brd 00.00.00.00.00.00

lines 127.0.0.1/8 scope host lo

valid 1ft forver preferred. 1ft forever

inets:: 112V Scope host

valid 1ft forever fruferenee 1ft forevers.

8) Ipconfig

The Ipconfig Command was a stragle in many sysadmins tool bet for configuration and trouble shooting network it has discused above.

output:

wipsso flag 4099 <UP, BROADCAST, MULTICAST> MN1500

either 16.5d : 7f : bd : 46:86: fregular 1000

ex packets 0 bytes 0 (00B)

ex error 0 dropped 0 overnins 0 name 0

TX Packets 0 bytes 0 (0.0B)

Tx errors 0 dropped overcome 0 carrier 0 collisions 0

9) mtr

MTR is a program with a command line interface that server as network diagonistic and trouble shooting tool

Output:

-F, --filename FILE read hostname (s) from

-U, --udp use UDP instead of icmp echo

-T, --tcp use TCP instead of icmp echo

-I, --para Port target Port number for tcp, SCTP.

-i --interval SECONDS ICMP echo request interval

10) Ping

Ping is a tool that verifies . IP-level connectivity to another TCP/IP by sending internet Control message Protocol Echo request message.

OUTPUT:

64 bytes from 130.46.149.34 b google usercontent.com

64 bytes from 130.46.149.31.bc - google usercelant. com (34.149.46.130)

icmp-seq=b ttl =120 time = 13.7ms.